

Communication Electronics Super Condensed Cheat Sheet (with Variable Definitions)

Waves & Antennas

- $\lambda = c/f$
(λ = wavelength [m], c = speed of light $\approx 3 \times 10^8$ m/s, f = frequency [Hz])
- Quarter-wave antenna:
 $L = \lambda/4$
(L = antenna length [m])
- Half-wave dipole:
 $L = \lambda/2$
- Propagation time:
 $t = d/v_p$
(t = time [s], d = distance [m], v_p = propagation velocity [m/s])
- Velocity in coax:
 $v = c \cdot VF$
(VF = velocity factor of coax, unitless fraction <1)

Decibels & Gains

- $G_v(dB) = 20 \log_{10}(V_{out}/V_{in})$
(V_{out} , V_{in} = output/input voltage [V])
- $G_i(dB) = 20 \log_{10}(I_{out}/I_{in})$
(I_{out} , I_{in} = current [A])
- $G_p(dB) = 10 \log_{10}(P_{out}/P_{in})$
(P_{out} , P_{in} = power [W])
- Cascaded:
 $G_{tot} = \prod A_n$
, in dB
 $G_{dB} = \sum G_{n,dB}$
(A = stage gain, dimensionless)
- $P_{dBm} = 10 \log_{10}(P/1mW)$
(P = power [W])

Tip: Multiply numeric gains, add dB. Loss is negative gain.

RC Filters

- $f_c = 1/(2\pi RC)$
(f_c = cutoff freq [Hz], R = resistance [Ω], C = capacitance [F])
- $H(j\omega) = \frac{\omega_c}{j\omega + \omega_c}$
(ω = angular freq = $2\pi f$ [rad/s])
- At f_c : -3 dB point,
 $|H(j\omega_c)| = 1/\sqrt{2}$
- Slope: -20 dB/decade/pole.

Resonance & RLC

- $X_L = 2\pi fL$
(X_L = inductive reactance [Ω], L = inductance [H])
- $X_C = 1/(2\pi fC)$
(X_C = capacitive reactance [Ω], C = capacitance [F])
- $f_0 = 1/(2\pi\sqrt{LC})$

(f0 = resonant freq [Hz])

- $Q = f_0 / BW$
- (Q = quality factor, BW = bandwidth [Hz])
- $BW = f_0 / Q$

Tip: At resonance, XL = XC. Tank circuits → high Z at f0.

Filters (Butterworth / Chebyshev)

- $L_{ref} = R_0 / (2\pi f_c), C_{ref} = 1 / (2\pi R_0 f_c)$
(R0 = reference resistance [Ω])
- Butterworth = flat passband, Chebyshev = ripple but sharper cutoff.

Matching Networks

- $Q = X_L / R_s$
(Rs = source resistance [Ω])
- $R_p = (Q^2 + 1) R_s$
(Rp = parallel resistance [Ω])
- $L_p = L_s (1 + 1 / Q^2)$
(Ls = series inductance [H])
- $C_p = 1 / ((2\pi f)^2 L_p)$
(Cp = capacitance [F])

Tip: Place shunt on higher impedance side.

Oscillators & AM

- Barkhausen:
 $|A\beta| \geq 1, \angle A\beta = 0^\circ$
(A = amp gain, β = feedback factor)
- AM:
 $s(t) = [1 + m \cos(\omega_m t)] \cos(\omega_c t)$
 - m = modulation index (Am/Ac), Am = message amplitude, Ac = carrier amplitude
 - ωm = modulating angular freq (2πfm), ωc = carrier angular freq (2πfc)

Tip: m>1 → overmodulation distortion.

High-Value Exam Problem Reminders:

- Antenna length ↔ frequency:** Use
 $f = c / \lambda$
. Then L=λ/4 or λ/2.
- Cascaded stages:** If dB → add, if numeric → multiply.
- Filter cutoff:**
 $f_c = 1 / (2\pi RC)$
. Slopes: -20 dB/decade per pole.
- Q-factor:**
 $Q = f_0 / BW$
(fast if f0, BW given).
- Power conversions:**
 $P_{dBm} = 10 \log_{10}(P / 1mW)$
↔
 $P(W) = 10^{(dBm/10)} / 1000$